

ML Syllabus Notes

Topic-wise Exam Revision

ML Syllabus Notes

1. Machine Learning kya hota hai?

- Machine Learning ek aisi technique hai jisme machine data se seekh kar decisions leti hai.
- Iska use prediction, classification, clustering aur pattern detection mein hota hai.
- ML ke main types hain:
 - Supervised Learning
 - Unsupervised Learning
 - Reinforcement Learning

2. Supervised Learning

- Isme model ko labeled data diya jata hai.
- Input aur output dono known hote hain.
- Main tasks:
 - Regression: continuous output
 - Classification: discrete output
- Common algorithms:
 - Linear Regression
 - Logistic Regression
 - Decision Tree
 - Random Forest
 - SVM
 - Naive Bayes
 - KNN

3. Unsupervised Learning

- Isme data labeled nahi hota.
- Model khud patterns dhoondhta hai.
- Main tasks:
 - Clustering
 - Association

- Dimensionality Reduction
- Common algorithms:
 - K-Means
 - Hierarchical Clustering
 - DBSCAN
 - PCA

4. Reinforcement Learning

- Isme agent environment mein action karta hai aur reward/punishment receive karta hai.
- Goal: maximum reward lana.
- Examples: robotics, gaming, self-driving systems.

5. Data Preprocessing

- Missing values handle karna
- Categorical data ko numeric mein convert karna
- Feature scaling: Standardization aur Normalization
- Train-test split karna
- Outliers ko handle karna
- Feature selection aur feature engineering

6. Linear Regression

- Continuous output ke liye use hota hai.
- Formula: $y = mx + c$
- Goal: best-fit line banana.
- Use cases: house price prediction, sales forecasting.
- Metrics: MSE, RMSE, MAE, R-squared

7. Logistic Regression

- Classification ke liye use hota hai.
- Output 0 ya 1 ke beech hota hai.
- Sigmoid function use hota hai.
- Use cases: spam detection, disease prediction.

8. Decision Tree

- Tree structure mein decision rules based on features.

- Easy to understand aur visualize.
- Problem: overfitting.
- Solution: pruning aur limiting depth.

9. Random Forest

- Multiple decision trees ka ensemble.
- Better accuracy aur less overfitting.
- Robust model hota hai.

10. Support Vector Machine (SVM)

- Classification aur regression dono mein use hota hai.
- Best boundary ya hyperplane find karta hai.
- Kernel functions use hote hain.

11. K-Nearest Neighbors (KNN)

- Prediction ke liye nearest neighbors ka use karta hai.
- Simple aur intuitive model hai.
- Large datasets mein slow ho sakta hai.

12. Naive Bayes

- Probabilistic classifier hai.
- Bayes theorem par based hai.
- Text classification mein popular hai.
- Example: spam filtering.

13. K-Means Clustering

- Unsupervised learning ka popular algorithm hai.
- Data ko K clusters mein divide karta hai.
- Steps:
 1. K centers choose karo
 2. Points ko nearest center mein assign karo
 3. Centers update karo
 4. Repeat karo

14. Hierarchical Clustering

- Data ko tree-like structure mein group karta hai.
- Agglomerative aur Divisive methods use hote hain.
- Small datasets mein useful hota hai.

15. Dimensionality Reduction

- Features ko reduce karke simpler model banana.
- PCA (Principal Component Analysis) common hai.
- Helps in visualization, speed aur overfitting control.

16. Model Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- ROC-AUC
- Cross-Validation

17. Overfitting aur Underfitting

- Overfitting: training data par achha, test data par weak
- Underfitting: model bahut simple hai aur dono par weak
- Solution: regularization, more data, feature selection, pruning

18. Bias-Variance Tradeoff

- Bias: model kitna simple hai
- Variance: data ke small changes par kitna react karta hai
- Best model balanced tradeoff provide karta hai

19. Neural Networks Basics

- Inspired by human brain.
- Neurons, weights, bias aur activation functions use hote hain.
- Deep Learning neural networks ka advanced form hai.

20. Important exam differences

- Regression vs Classification
- Supervised vs Unsupervised
- Clustering vs Classification
- Precision vs Recall
- Bias vs Variance
- Overfitting vs Underfitting

21. Important exam tips

- Definitions yaad rakho
- Algorithms ke use cases yaad rakho
- Confusion matrix aur metrics practice karo
- Comparative differences likh kar revise karo
- Short notes aur flowcharts bana lo

22. Short revision summary

- Supervised learning: labeled data
- Unsupervised learning: unlabeled data
- Regression: continuous output
- Classification: discrete output
- Clustering: similar data ko group karna
- Evaluation metrics decide karte hain model kitna accha hai

Prepared for exam revision